

SOFIANE MALLEM

PYTHON SOFTWARE DEVELOPER

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SUMMARY

Python Developer with professional experience building and operating production Python systems, including horizontally scaled rendering infrastructure, automation tooling, and data pipelines. Owned a CAD-to-rendering workflow spanning customer data ingestion, distributed rendering, and downstream ML training, delivering measurable improvements in performance and operational reliability.

EXPERIENCE

Python Developer, Synthavo

Nov 2025 to Apr 2026

Stuttgart, BW

- Owned the customer CAD data pipeline, validating, transforming, and uploading incoming datasets powering a horizontally scaled Blender rendering service across 5+ Linux servers.
- Migrated the rendering service to a newer Blender version, improving render speed by a further ~40% on top of prior optimization work, while sustaining over 99% uptime through proactive monitoring and bottleneck resolution.
- Maintained the production rendering-to-training pipeline, ensuring reliable transfer of generated datasets into YOLOv5 model training workflows and coordinating delivery of trained models to the integration and QA teams.
- Acted as primary technical contact for production incidents across the pipeline, diagnosing failures and restoring service with minimal disruption.

Python Developer Intern, Synthavo

Feb 2025 to Aug 2025

Stuttgart, BW

- Developed Python tooling supporting the CAD validation and transformation pipeline, improving reliability of production rendering workflows.
- Applied mesh reduction to incoming STL files ahead of rendering, improving render throughput by approximately 40%.
- Improved customer onboarding workflow efficiency, cutting provisioning time from ~5 days to 2–3 days through Python tooling and process automation.

PROJECTS

Durden — YOLO Object Detection Service

Stack: FastAPI · YOLO · Docker · GitHub Actions · GHCR · Google Cloud Run · Workload Identity Federation

- Built and deployed an object detection web app with live confidence scoring and runtime selection across multiple YOLO models.
- Implemented full CI/CD in GitHub Actions: linting, unit tests, image build, and publish to GHCR on every push.
- Deployed to Google Cloud Run using Workload Identity Federation, enabling keyless GitHub Actions authentication with no long-lived credentials.
- Configured autoscaling (0–1 instances) for cost efficiency, with unit tests gating every deployment.

Hang — Scalable Distributed Messaging Platform

Stack: FastAPI · WebSockets · Redis Pub/Sub · PostgreSQL · Docker · Nginx · Google OAuth · S3 · Prometheus

- Designed and developed a distributed real-time messaging platform supporting authenticated users, persistent messaging, typing indicators, read receipts, and media uploads.
- Architected a horizontally scalable WebSocket layer using Redis Pub/Sub to synchronize events across multiple stateless instances.
- Deployed backend nodes behind an Nginx load balancer via Docker, enabling horizontal scaling without application changes.
- Implemented cursor-based pagination with lazy loading to efficiently serve large conversation histories while minimizing database load.
- Integrated Google OAuth and S3-compatible object storage for identity management and media handling.
- Provisioned and load-tested a complete local production-like environment using Docker, PostgreSQL, Redis, and Prometheus for multi-service integration testing.

EDUCATION

Software Engineering Course

Feb 2023 to Nov 2025

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Bachelor of Computer Science

2017 to 2020

University of S.T Houari Boumedien, Algiers, Algeria

TECHNICAL PROFICIENCIES

Programming: Python, SQL, JavaScript, C, C++, Bash

Backend Development: FastAPI, REST APIs, API Design, WebSockets, OAuth, Async Python

Databases & Data: PostgreSQL, Redis, Pandas, Data Pipelines, CAD Processing

Infrastructure & DevOps: Distributed Systems, Docker, CI/CD, GitHub Actions, Google Cloud, Linux, Nginx, Prometheus

Development Practices: Agile, Git, Pytest, Unit Testing, Integration Testing

LANGUAGES

English (Fluent), French (Native), Arabic (Native), German (Basic)

LINKS

<https://www.linkedin.com/in/sofiane-mallem-b87302170>, <https://github.com/Melsso>
